

Use It or Lose It?

Programmable money and the digital euro's interest rate

Speaker script · 20 minutes + 10 minutes Q&A

The story in twelve sentences (the slide titles)

1. Use it or lose it?
2. Money with an expiry date gets spent, and fast.
3. A digital euro could build the same pressure into money itself, by an expiry date or by continuous decay.
4. To compare them, take a holder who searches for good uses of money and can always dump it at a loss.
5. An expiry date pushes out more spending, but ends in a fire sale and splits money into vintages.
6. Continuous decay keeps every euro equal, and it is nothing more than a negative interest rate.
7. As a rate, decay is far too weak to replicate a voucher: money can carry a rate, not a stimulus programme.
8. So the real question is the digital euro's interest rate, and the draft regulation fixes it at zero.
9. A zero-rate digital euro is cash without storage costs, which lifts the lower bound back to zero.
10. A negative digital-euro rate can win that space back, but no more while cash still exists.
11. The best design keeps zero as the default and lets balances above a threshold decay when policy rates turn negative.
12. The digital euro should be able to bear interest: only as decay, and only when rates are negative.

Timing

1	Title	0:30	7	Too weak for a voucher	1:45
2	Expiry gets spent	1:30	8	Art. 16(8)	1:15
3	Two ways to program	1:30	9	Cash without storage costs	2:30
4	Model	2:00	10	Win back, no more	2:00
5	Expiry results	2:00	11	Contingent clause	2:00
6	Decay is a rate	2:00	12	Answer	1:00

Checkpoints: slide 4 done by 5:30, slide 7 done by 11:15, slide 10 done by 16:00.

Numbers to know

- China coupons: about ¥3 of extra spending per ¥1 of subsidy. France: one-month MPC 61% with a three-week expiry, 23% without.
- Art. 16(8): “The digital euro shall not bear interest.” Holding limit under discussion: €1,500–3,000.
- ECB deposit facility rate: –0.50%, September 2019 to July 2022.
- 30-day comparison: useful spending 84% (expiry) vs 73% (decay); 16% dumped at the deadline.
- Decay needed to match a voucher: 160%–1,200% a year. A 1% decay raises velocity by about 12%.

Slide 1

0:30

Use It or Lose It?

Programmable money and the digital euro's interest rate

[Presenter name]

October 2026

1

Say

- “Governments have discovered that if you give people money with a deadline, they spend it.”
- “Today I ask whether a central bank should build that idea into money itself, and what it means for the digital euro.”

Slide 2

1:30

Money with an expiry date gets spent, and fast

China, digital coupons, 2020

¥3

extra spending per ¥1 of subsidy
coupons valid for about one week

France, randomised prepaid cards

61% vs 23%

one-month MPC, with vs without
a three-week expiry

A deadline turns a transfer that would be saved into spending *now*.

Xing, Zou, Yin, Wang & Li (AEJ: Macro 2023); Ding, Jiang, Msall & Notowidigdo (2024); Boehm, Fize & Jaravel (2024).

2

Say

- In 2020, Chinese cities handed out digital coupons through payment apps, valid for about a week. Each yuan of subsidy produced about three yuan of extra spending.
- In France, a randomised experiment gave households prepaid cards. With a three-week expiry, 61% was spent within a month; without one, 23%.
- “The deadline is what turns a transfer that would be saved into spending today.”

If asked

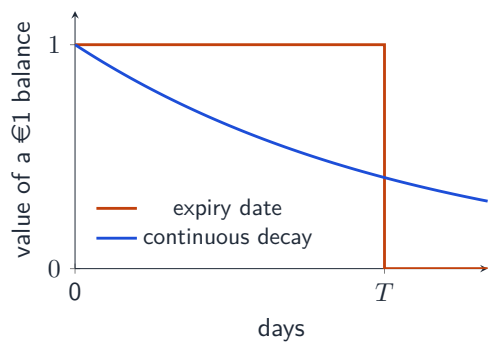
Q *Weren't the Chinese coupons driven by minimum-spend thresholds, not expiry?*

A Partly. Ding et al. point to the thresholds. That is why I show the French experiment, which isolates the deadline.

Slide 3

1:30

A digital euro could build the same pressure into money itself, by an expiry date or by continuous decay



- In a liquidity trap, money is hoarded and velocity falls
- A programmable currency can charge for holding it
- **Expiry:** the whole cost falls on day T
- **Decay (demurrage):** the same cost spread evenly over time

Which of the two should a central bank use?

3

Say

- In a liquidity trap, the central bank adds money and people sit on it; velocity falls.
- A programmable digital euro could charge people for sitting on money, in two ways.
- An expiry date: nothing happens until day T , then the money is gone.
- Continuous decay, what Gesell called demurrage: the balance shrinks a little every day.
- “Same idea, different timing. Which should a central bank use?”

Slide 4

2:00

To compare them, take a holder who searches for good uses of money and can always dump it at a loss

- A unit of money is worth V to its holder; a good purchase is worth 1
- Searching at intensity κ costs $\kappa^2/2a$; κ is the **velocity**
- A bad use (junk goods, selling at a discount) is always available at $\phi < 1$

Expiry at T ($\tau = T - t$ left)

$$\frac{dV}{d\tau} = -rV + \frac{a}{2}(1 - V)^2, \quad V(0) = \phi$$

Decay at rate λ

$$(r + \lambda)V = \frac{a}{2}(1 - V)^2$$

In both cases the holder spends at speed $\kappa^* = a(1 - V)$: the less money is worth holding, the faster it moves.

4

Say

- A deliberately simple model. Each unit of money is worth V to the holder. A good purchase is worth 1, but finding it takes effort.
- The search speed κ is the velocity; it costs $\kappa^2/2a$.
- There is always a bad option: junk goods, or selling the token at a discount, worth ϕ .
- With an expiry date, value depends on time left and ends at ϕ . With decay, the decay rate simply adds to the discount rate.
- “In both cases, the less money is worth holding, the faster it is spent.”

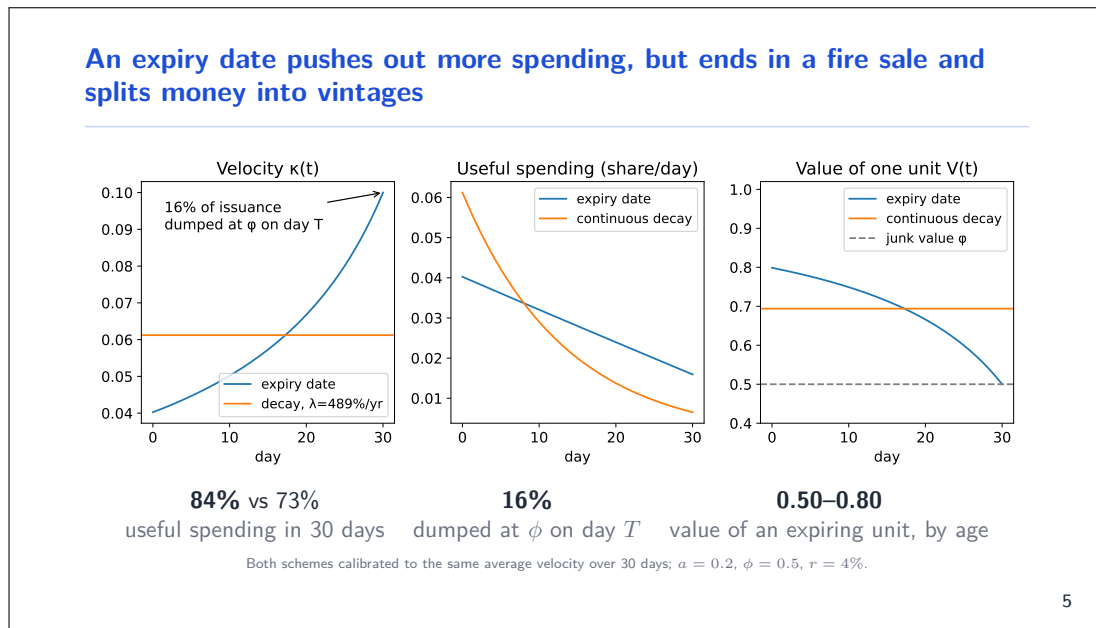
If asked

Q Why a quadratic search cost?

A It is the simplest convex cost with an interior solution. With linear cost, velocity would literally explode at the deadline; the convex case is the conservative one.

Slide 5

2:00



Say

- I tune the decay rate so both schemes have the same average velocity over 30 days.
- Left: under expiry, spending accelerates into the deadline; under decay it is flat.
- The expiry date actually gets more useful spending done: 84% against 73%. It forces the whole balance out.
- But 16% is still unspent on the last day and is dumped on junk.
- Right: an expiring token's value depends on its age. A fresh one is worth 0.8, an old one 0.5. Two euros are no longer the same euro.

If asked

Q *Why match average velocity?*

A Velocity is the target in a liquidity trap. Matching total spending is impossible here: decay never reaches the expiry level, which is itself informative.

Slide 6

2:00

Continuous decay keeps every euro equal, and it is nothing more than a negative interest rate

	Expiry	Decay
Fire sale at the deadline	3–55%	none
Value of one unit	by vintage	all equal
One euro = one euro?	no	yes
Applies to all money	no	yes

Decay at λ is the rate $i = -\lambda$.
Velocity follows

$$\kappa^* \approx \sqrt{2a(r + \lambda)}$$

the square-root rule of Baumol–Tobin money demand.

Expiry is a feature of a spending programme. Decay can be a feature of money.

6

Say

- Put the two side by side. Expiry ends in a fire sale, prices each unit by its age, and so breaks the rule that one euro equals one euro.
- It also cannot apply to all money; it needs a separate vintage for every payment.
- Decay has none of these problems. And decay at rate λ is exactly an interest rate of $-\lambda$.
- Velocity responds with the square root of the holding cost, the same rule as Baumol–Tobin money demand.
- “An expiry date is a feature of a spending programme. Decay can be a feature of money.”

If asked

Q *Couldn't a smart contract track the vintages?*

A Technically yes. But then every wallet holds euros of different value, and shops must either take them at par, which invites arbitrage, or at a discount, which ends the singleness of money.

Slide 7

1:45

As a rate, decay is far too weak to replicate a voucher: money can carry a rate, not a stimulus programme

Decay needed to match a voucher

160–1,200%

per year (90-day to 7-day vouchers)

Policy-sized rates, -0.5% to -3%

+6–32%

velocity

- Vouchers also work through new money, a salient deadline and minimum-spend thresholds
- None of these can be a permanent property of a currency

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Say

- How strong would decay need to be to act like a voucher? Between 160% and 1,200% a year.
- Policy-sized negative rates, half a percent to three percent, raise velocity by 6 to 32%.
- Vouchers also rely on things a currency cannot have permanently: new money, a salient deadline, spending thresholds.
- “So money cannot be a stimulus programme. What it can carry is an interest rate.”

If asked

Q *Don't these numbers depend on your parameters?*

A The levels do; the order of magnitude does not. Across the sensitivity grid the matching rate stays in the hundreds of percent (backup B3).

Slide 8

1:15

So the real question is the digital euro's interest rate, and the draft regulation fixes it at zero

“The digital euro shall not bear interest.”

Commission proposal for a digital euro regulation, Art. 16(8), June 2023

- Not positive, not negative, in any circumstances
- Paired with a holding limit, currently discussed at €1,500–3,000 per person
- Council and Parliament are negotiating the final text now

8

Say

- “That brings us to the digital euro, whose rate is being decided right now.”
- The Commission’s draft settles it in one sentence: the digital euro shall not bear interest. Not positive, not negative, ever.
- It comes with a holding limit, around €1,500 to 3,000 per person. Council and Parliament are negotiating the final text.

Slide 9

2:30

A zero-rate digital euro is cash without storage costs, which lifts the lower bound back to zero

Depositors can always switch to central-bank money, so banks cannot pay less than

$$r_D \geq \max_j \{i_j - c_j\} \quad (i_j: \text{rate}, c_j: \text{cost of holding})$$

	rate	holding cost	floor for deposit rates
Cash	0	storage, insurance $c > 0$	$-c$
Digital euro at zero	0	≈ 0	0

- 2019–22: the ECB went to -0.50% because hoarding banknotes is costly
- Next time, every holder can move up to the limit into a costless zero-rate asset
- Deposits leave banks exactly when negative rates already squeeze their margins

9

Say

- Depositors can always leave the bank for central-bank money. So a bank cannot pay less than the best such option, net of what it costs to hold it.
- Cash pays zero but is costly to store and insure. That is why the ECB could go to -0.50% in 2019–22 without people emptying their accounts.
- A zero-rate digital euro is cash without those costs. Up to the limit, it puts the floor back at exactly zero.
- And the outflow comes precisely when negative rates are already squeezing bank margins.

If asked

Q *Retail deposit rates never went negative anyway.*

A For small balances, banks chose not to charge. Larger and corporate deposits did pay negative rates. A zero-rate digital euro turns that choice into a legal floor for everyone up to the limit.

Q *What about stablecoins?*

A Under MiCA, e-money tokens cannot pay interest either. They are another zero-rate option, but with issuer risk.

Slide 10

2:00

A negative digital-euro rate can win that space back, but no more while cash still exists

$$\text{floor} = \underbrace{\max\{0, -c\}}_{\text{zero rate}} = 0 \quad \longrightarrow \quad \underbrace{\max\{-\lambda, -c\}}_{\text{decay at } \lambda} \geq -c$$

- A zero rate **raises** the floor from $-c$ to 0
- Decay can bring it back down to $-c$, but **not below**: beyond that, people switch to banknotes
- Going deeper requires repricing cash itself, e.g. a crawling cash–reserve exchange rate

A flexible rate protects the policy space we have; it cannot create more.

Cash repricing: Agarwal & Kimball (IMF WP 19/84, 2019).

10

Say

- If the digital euro could decay, the floor becomes the higher of $-\lambda$ and the cash floor.
- So decay wins back the space that the zero rate gave away, down to $-c$.
- But not further: below $-c$, people simply switch to banknotes.
- Going deeper would need cash itself to be repriced, for instance Agarwal and Kimball's crawling exchange rate between cash and reserves.
- “A flexible rate protects the policy space we have. It cannot create more.”

If asked

Q *How large is c ?*

A It is not estimated here. The 2019–22 episode shows it is at least as large as the negative rates passed on to large depositors without a run into cash.

Q *Isn't this just Bordo and Levin (2017)?*

A They show an interest-bearing CBDC could remove the lower bound. With cash kept, as in the euro area, a digital-euro rate alone can only restore the cash floor, not remove it.

Slide 11

2:00

The best design keeps zero as the default and lets balances above a threshold decay when policy rates turn negative

A contingent rate clause

$$i_{DE} = \begin{cases} 0 & \text{on all balances, when DFR} \geq 0 \\ 0 & \text{on balances up to } M_1 \\ \max\{\text{DFR}, -\bar{c}\} & \text{on balances above } M_1, \text{ when DFR} < 0 \end{cases}$$

- **Zero in normal times:** simple, trusted, never “money that melts”
- **Small balances never decay:** households and payments are protected
- **Decay, never expiry:** every euro stays equal to every other
- **Floored at $-\bar{c}$:** no reason to flee into banknotes

11

Say

- The design follows directly. In normal times: zero, as in the draft. Simple and trusted.
- Only when the policy rate is negative do balances above a threshold M_1 decay, at the policy rate, but never below the cost of holding cash.
- Small balances never decay, so households and everyday payments are untouched.
- Decay, never expiry, so every euro stays equal to every other.

If asked

Q *Who would hold a digital euro that can shrink?*

A Nobody's first M_1 ever shrinks. Above it, the charge equals what a large bank deposit would pay; the digital euro is no worse than a bank account, it just stops being better.

Q *The ECB says the digital euro will not be programmable money.*

A That means no restrictions on what it can buy. Decay restricts nothing; it is a rate on a balance, like interest on reserves.

Slide 12

1:00

The digital euro should be able to bear interest: only as decay, and only when rates are negative

1. Deadlines make people spend, but an expiry date belongs to a spending programme, not to money
2. The only holding cost money can carry is continuous decay, which is simply a negative interest rate
3. Fixing that rate at zero turns the digital euro into costless cash and gives up policy space
4. A contingent, tiered rate keeps zero for normal times and protects that space when it matters

12

Say

- Read the four points.
- Close: “Deadlines belong to spending programmes. Money can only carry a rate. The digital euro should be allowed to carry one: zero in normal times, decay when rates are negative, and never an expiry date.”

More questions

Q *After the 2022 inflation, why worry about negative rates?*

A The lower bound bound for most of 2009–2021. A regulation lasts decades; it should not be designed for the current regime only.

Q *Your model is partial equilibrium.*

A Yes. The qualitative results, that expiry breaks fungibility and that decay is a rate, do not depend on general equilibrium.

Q *Do coupons just pull spending forward?*

A Evidence is mixed: Xing et al. (2023) find purchases moved forward, Ding et al. (2024) do not. The argument here does not rely on it.

Q *Wörgl's stamp scrip was banned. Why would demurrage be acceptable now?*

A It was banned because a town issued money, not because demurrage failed. Here the issuer is the central bank and the charge is a policy rate.

Q *Does decay hurt the poor?*

A Balances below M_1 are exempt, which covers most low-income holdings.

Q *Is this within the ECB's mandate?*

A Setting the rate on its own liabilities is standard monetary policy; Art. 16(8) would need a conditional clause instead of a blanket ban.

Glossary

Velocity	Spending per unit of money per period, $V = PY/M$.
MPC	Share of an extra euro spent within a period.
Liquidity trap	More money does not raise spending because it is hoarded.
Demurrage / decay	A continuous charge on holding money; a negative interest rate.
Singleness of money	Every form of a currency exchanges one-for-one at par.
Fungibility	Any unit is a perfect substitute for any other.
Baumol–Tobin	Inventory model of money demand with interest elasticity $-\frac{1}{2}$.
ZLB / ELB	Zero / effective lower bound on nominal interest rates.
DFR	ECB deposit facility rate, its main policy rate.
Tiered remuneration	Different rates on different slices of a balance.
Holding limit	Maximum digital-euro balance per person; the excess is swept to a bank account.
Art. 16(8)	“The digital euro shall not bear interest.”
MiCA	EU crypto-asset regulation; bans interest on e-money tokens.
